CE-BPSO-AHDL-IDS: Compute-Efficient Binary Particle Swarm Optimization-Based Architecture Search for Hybrid Deep Learning Intrusion Detection in the Internet of Things

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***Abstract*—Neural architecture search for intrusion detection requires useful evaluation under limited compute. This project retains a 73-bit BPSO CNN–RNN framework and investigates multi-fidelity allocation on real Gas Pipeline data. An earlier archive-based uncertainty-aware method reduced online GPU use but required 403.82 GPU seconds of predictor observations, eliminating end-to-end savings. We implement an archive-free candidate that begins as BPSO with successive halving and learns only from search-generated observations using architecture-disjoint, prequentially checked CPU prediction. In 42 fresh bounded runs across three seeds, archive-free full used 11.947 GPU seconds on average to reach validation macro-F1 target 0.6 or exhaust the cap, versus 16.824 for standard BPSO: 28.99% less total GPU computation. Targets were attained in 2/3 versus 1/3 seeds. However, predictive promotion never activated in the short searches; results matched ordinary halving, and the measured saving is attributable to multi-fidelity fallback, not online uncertainty learning. DEHB reached 3/3 targets using 6.778 GPU seconds. All failed target runs were retained as censored. Existing test results use a development-known historical split. The work supports reproducible engineering and removal of archive overhead; superiority or novelty of uncertainty-aware optimization remains unsupported.**

***Index Terms*—Intrusion detection, neural architecture search, binary particle swarm optimization, training budgets, class imbalance.**

I. Introduction

An intrusion detector needs to identify rare attacks without generating too many false alarms on normal traffic. Building a hybrid neural model also means choosing its depth, recurrent units, optimizer and training duration. Architecture search explores these choices, but each candidate has to be trained. Reducing that cost with shorter runs is useful only when the resulting scores still distinguish promising candidates.

We use BPSO-AHDL-IDS, the convolutional and recurrent architecture-search framework proposed by Lu et al. [1], to study model selection under a limited training budget. The research question was whether short-budget fitness provided useful evidence about later validation performance after ordinary learning and preprocessing controls were examined. If a model predicts only one class, its accuracy can be misleading and different architectures can receive the same fitness. We therefore check the learning setup and preprocessing before treating such results as a reason to change the search algorithm.

The contributions supported by this study are: (1) an executable budget-constrained BPSO framework with explicit reproduction choices and validation-only search; (2) a repeated-seed, four-preset study of 3/10/30-epoch fitness and selected 200-epoch learning trajectories; (3) normalization, neural-component and predictor-state grouping controls on real Gas data; and (4) a fixed 200-epoch reference evaluation at three dedicated final-training seeds. The follow-up adds FP32 execution optimization, target-stopping and actual-GPU-budget experiments. The evidence supports empirical and reproducibility contributions. A separate candidate extension adds architecture-disjoint learning-curve prediction and uncertainty-aware resource allocation, with bounded matched-time comparisons and ablations (Section VI). No novel BPSO update or established algorithmic novelty is claimed.

II. Related Work

The source method [1] represents network choices as binary strings and uses BPSO to search for a hybrid CNN/recurrent model. It reported Gas accuracy 0.9897 and F1 0.9744. These values belong to the source study and are included only as reference measurements. Unknown source-paper partition identities, seeds, feature ordering and multiclass averaging prevent a controlled numerical reproduction claim.

We include random search [2] as a baseline for selecting architectures without a guided proposal rule. Successive halving [3] gives larger budgets to candidates that survive each rung, and Hyperband [4] runs several such brackets. DEHB [5] adds differential-evolution proposals to Hyperband; BOHB [6] uses model-based proposals with multi-fidelity allocation. These existing methods make adaptive budgeting a baseline requirement rather than sufficient evidence of novelty. We did not implement BOHB.

The question for this IDS search space is whether low-budget scores reject candidates that improve with more training. We also ask whether a different allocation rule gives better selections for the same resources. These are questions about the tested search space, not assumptions about architecture search in general. The original Gas documentation [7], [8] also motivates explicit label verification and scrutiny of operating-state dependence.

III. Problem Formulation and Methodology

A. Budget-Constrained Selection

For architecture a, training seed s and cumulative epoch budget b, define

$$f_b(a, s) = \mathrm{F1}_{\mathrm{macro}}(D_{\mathrm{va}}; \mathrm{Train}(a, D_{\mathrm{tr}}, b, s)). \tag{1}$$

The search maximizes observed validation fitness under an experiment cap. For evaluation requests q with incremental completed epochs Δe_q, the resource-limited form is

$$\max_a f_B(a, s) \quad \text{subject to} \quad \sum_q \Delta e_q \le C_e. \tag{2}$$

Resource caps can bound search wall time or measured GPU training/validation elapsed time. The fixed-quality protocol stops only after an actual 30-epoch target is reached; fixed-compute comparisons use common GPU horizons. Epochs do not equal FLOPs across different models. An equal cap also need not imply equal consumption, because a full promotion may not fit the remaining allowance. Actual epochs, selected fidelity, elapsed search time and failures must therefore accompany a method comparison.

Search uses training and validation only. Architecture, preprocessing, loss and training duration are fixed before final evaluation. Following the validation studies, the reference hybrid was independently initialized at three dedicated training seeds and evaluated on the existing test split after exactly 200 epochs. This fixed-reference evaluation is separate from BPSO candidate selection and from the earlier validation continuations.

B. Hybrid Model and Encoding

Each record (N, F) is reshaped to $(N, 1, F)$ for one-dimensional convolution, activation and max pooling. The tensor is then transposed to (N, L, C) for sequence-preserving RNN, LSTM, GRU or linear blocks. A classifier reads the last sequence position. This is feature-axis processing, not a temporal window of successive events. Encoded recurrent activations follow each module; native PyTorch recurrent gates are retained.

The paper-informed codec has 73 bits: eight for batch size, optimizer and learning rate, two for CNN count, four ten-bit CNN slots, one for recurrent count, and two eleven-bit recurrent slots. Cache keys use only the active parts of the decoded architecture. Fixed-slot count mappings, pooling, output aggregation and feature order were explicit implementation choices. When a short feature sequence would make an operation invalid, the implementation pads it and records the repair. This policy and the encoding are documented choices; the original authors' code was unavailable for an exact comparison.

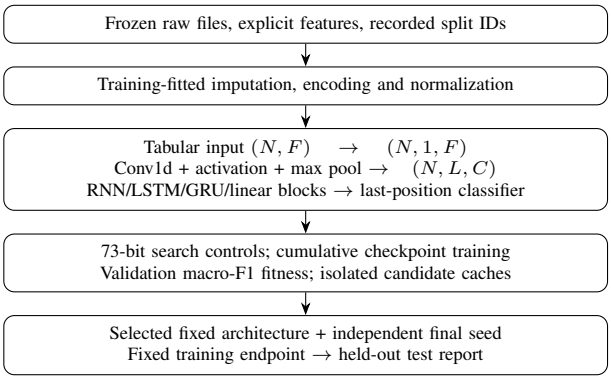


Fig. 1. Framework architecture. Recurrent blocks traverse the transformed feature axis of one tabular record. Test data are used after the architecture and training endpoint have been fixed.

C. Training and Swarm Update

The minibatch objective is

$$\mathcal{L} = -\frac{\sum_{i \in B} \alpha_{y_i} \log p_\theta(y_i \mid x_i)}{\sum_{i \in B} \alpha_{y_i}} + \lambda_1 \sum_{W \in \mathcal{W}_r^{(1)}} \|W\|_1 + \lambda_2 \sum_{W \in \mathcal{W}_r^{(2)}} \|W\|_F^2. \tag{3}$$

Balanced weights are proportional to inverse training-class frequency, normalized to unit mean. Validation and test frequencies do not fit them. The encoding applies L1/L2 penalties to recurrent weights, although the preset controls evaluated here did not enable them. The framework defaults to unweighted loss; balanced loss was set explicitly for this study.

With x_{ij} denoting the particle position, p_{ij} its personal best and g_j the global best, the BPSO update is

$$w_t = w_{\max} - (w_{\max} - w_{\min})t/G, \tag{4}$$

$$v'_{ij} = w_t v_{ij} + c_1 r_{1,ij}(p_{ij} - x_{ij}) + c_2 r_{2,ij}(g_j - x_{ij}), \tag{5}$$

$$x'_{ij} = \mathbf{1}\{r_{ij} \le \sigma(v_{ij})\}. \tag{6}$$

The function σ is the logistic sigmoid; each random factor is sampled independently. We interpret the source equation by replacing an out-of-bound velocity with a fresh Uniform$[0, 1]$ draw before updating the position. A best record changes only when validation macro-F1 strictly improves. We treat the conflicting global-best inequality in the printed source as a typographical error. The search returns candidates that were actually evaluated.

Algorithm 1: cumulative-budget search.

1) Record the raw-file checksums, split IDs, training transforms, codec, seeds and compute limits before starting.
2) Start the scheduler and request a candidate configuration.
3) Decode the architecture and resume its matching seed/data/configuration checkpoint, or initialize it if no checkpoint exists.

TABLE I
VERIFIED GAS PIPELINE CLASS NAMES. ORIGINAL NUMERIC TARGETS WERE PRESERVED.

Class (code)	Meaning
Normal (0)	Normal operation
NMRI (1)	Naive malicious response injection
CMRI (2)	Complex malicious response injection
MSCI (3)	Malicious state command injection
MPCI (4)	Malicious parameter command injection
MFCI (5)	Malicious function command injection
DoS (6)	Denial of service
Recon (7)	Reconnaissance

4) Train to the requested cumulative budget, preserve optimizer/RNG state, and score validation macro-F1.
5) Record incremental resource use, update the scheduler and promote or propose candidates until the schedule or cap ends.
6) Record the best observed candidate and its fidelity. Evaluate a separately frozen final model on locked test data only when that experiment is performed.

The seven implemented paths are BPSO, reduced BPSO, BPSO with successive halving, random, Hyperband, upstream DEHB 0.1.1 and a conventional random-forest proposal control. In BPSO with halving, only fitness measured at the final rung can update the swarm bests. Paper-informed Gas settings specify 20 particles, 30 populations, 30 candidate epochs and 200 final epochs, with $c_1 = c_2 = 1.5$, inertia settings 0.8/0.4 and velocity bounds ± 10. The full-scale method comparison at these settings has not yet been run.

IV. DATA AND EXPERIMENTAL PROTOCOL

A. Dataset and Preprocessing

The Gas ARFF obtained through the dataset author's link contained 97,019 records with 26 numeric predictors and eight target codes. The source hash and feature allowlist were preserved. The code meanings in Table I follow the dataset documentation [7], [8]. Stratified splitting with seeds 1729/1730 gave 58,211 training records, 19,404 validation records and 19,404 test records. We fixed these partitions for this project; they do not recover the row identities used in the source paper.

Imputation and normalization used training data only, and alternate target columns were removed. The column L2, row L2 and Min–Max experiments shared the same training/validation rows and fitted imputer. Column L2 divides a feature by its training-column norm, replacing a zero norm with one. Row L2 divides each record by its own norm. Min–Max parameters used training extrema. None of these transformations fitted validation extrema.

The original source defines `time` as a Modbus query–response interval [7], not an acquisition timestamp. Sorting this field cannot establish chronology. Original ToN-IoT, Bot-IoT and SWaT data were unavailable; generated profiles served only as software fixtures and supplied no original-benchmark efficacy evidence.

B. Preset Cohort and Neural Controls

The repeated-seed study froze four hybrid presets before reading new scores. Each used Adam at 0.001, batch 256, balanced cross entropy, clipping norm 1 and GRU dropout 0.1. Small, reference and wide presets used CNN/GRU widths 8/16, 16/32 and 32/64. Their CNN kernel was 3 and pool size 2. The stacked preset added a CNN32 block with pool size 1 to the reference CNN16 block, followed by GRU32. The cohort was a controlled four-preset sensitivity study, not a sample of the full 73-bit search space.

All four presets reached 3, 10 and 30 epochs under column L2 for seeds 42, 43 and 44. The reference also reached these budgets under Min–Max and row L2. The additional Min–Max models were MLP64/32, CNN16 with a flattened 32-unit head, and GRU32 over the feature sequence. We used them as conventional model-family controls. Since their widths, readouts and parameter counts differ, their scores cannot isolate the effect of one component at equal capacity.

Before observing the later scores, we specified continuations to epoch 200 for all three column-L2 reference seeds and to 200 for the Min–Max reference at seed 42. Each of these four continuations resumed its existing weights. They are extensions of earlier runs, not four additional independent fits. In total, the extension contained 27 neural trajectories through epoch 30 and four longer continuations.

C. Predictor-State Grouping

No exact predictor overlap was found across the original random partitions when all 26 features were included. After excluding time, the audit found 2,033 hashes shared by training and validation and 2,087 shared by training and test. Across the full corpus, 105 non-time predictor hashes had conflicting labels. This identifies repeated states, but it does not measure how much they affect the scores.

A separate sensitivity split grouped identical 25-feature non-time states using predictor-only hashes, without labels. Seeded group splitting reserved 20% of groups and split the remaining pool into training and validation groups. It produced 62,198 training, 14,462 validation and 20,359 reserved rows across 7,675 unique groups, with zero shared groups between partitions. Every class occurred in training and validation. Only training/validation arrays were made available to the new runs.

This split evaluated balanced 100-tree random forests, balanced logistic regression with training-fitted standard scaling, and the reference hybrid for seeds 42/43/44. The hybrid used training-fitted Min–Max and 30 epochs. Grouping separates exact predictor states in this release. Capture identities and actual timestamps are unavailable, so near-duplicate records may still be dependent. The grouped and random experiments also have different row counts, class proportions and hardware. Their score difference therefore cannot be used to measure the effect of leakage alone.

TABLE II
VALIDATION MACRO-F1, MEAN ± SAMPLE SD OVER THREE TRAINING SEEDS, COMPLETE FOUR-PRESET COLUMN-L2 COHORT.

Preset	Epoch 3 F1	Epoch 10 F1	Epoch 30 F1
Small	0.0460 ± 0.0460	0.0794 ± 0.0632	0.1724 ± 0.0478
Reference	0.0243 ± 0.0094	0.1309 ± 0.1186	0.4267 ± 0.2074
Wide	0.0705 ± 0.0550	0.4912 ± 0.0715	0.7348 ± 0.0868
Stacked	0.0194 ± 0.0142	0.2903 ± 0.2629	0.4735 ± 0.1835

D. Metrics, Repeats and Hardware

Macro-F1 includes all eight training classes, with zero assigned when a class metric is undefined. We also report accuracy, per-class recall, predicted-class coverage, confusion matrices and the normal-versus-attack false-positive rate (FPR). The majority baseline predicts the most frequent training label. Means and sample standard deviations summarize three seeds on the same partition; they describe training variability, not variation across independently collected datasets.

For each seed, we compared the same four candidates using Spearman/Kendall correlation and selection regret, $\max_a f_{30}(a) - f_{30}(\arg\max_a f_b(a))$. We left statistics undefined for constant rankings and reported regret intervals when selection-boundary ties prevented a unique choice. Four candidates are too few to characterize the full search space.

The random-split neural runs and final reference evaluations used an RTX 3050 Laptop GPU with 4 GB, Python 3.14.4, PyTorch 2.15.0 and CUDA 13.2. We ran the state-grouped hybrid controls on CPU and saved the environment and timing records with each experiment. An epoch counts a training pass, not a fixed amount of computation. Synchronized GPU seconds measure elapsed occupancy rather than energy. The validation-study directories contained no test arrays. For final evaluation, we reused the original test partition, whose earlier outcomes were already known; this was neither a new blind test nor an independent-release evaluation.

E. Fixed-Endpoint Reference Evaluation

The CNN16–GRU32 reference used training-fitted Min–Max scaling, balanced cross entropy, Adam at learning rate 0.001, batch size 256, dropout 0.1 and gradient clipping at norm 1. Before reading new test predictions, we fixed seeds 10042, 10043 and 10044 and required 200 epochs for each run. The models were initialized independently. Validation peaks were not used to select checkpoints or stop training. We checked the source checksum and all partition row identities against the original preparation, then generated test predictions only at the fixed endpoint and re-scored them independently. These runs measure training variability of the fixed reference on the existing split. A full-budget BPSO search did not select this model.

V. RESULTS

A. Training-Budget Sensitivity

Tables II and III report the complete cohort. For every seed, the best preset at epoch 10 was also best at epoch 30. Selection regret was zero, and Spearman correlation was 0.8. At epoch 3, only seed 43 had positive regret, equal to 0.0284. The ordering of other presets changed, but we did not observe a large failure to retain the later winner. This finding applies to the four tested presets and budgets; it does not show that a new allocation rule would improve selection. No selected 200-epoch subset was used to infer an unobserved full-cohort ranking.

TABLE III
WITHIN-SEED EARLY-TO-30-EPOCH RANKINGS FOR FOUR PRESETS. DASHES DENOTE NON-IDENTIFIABLE STATISTICS, NOT ZERO.

Seed	Early epoch	Spearman	Kendall	Regret
42	3	0.9487	0.9129	0.0000
42	10	0.8000	0.6667	0.0000
43	3	0.3162	0.1826	0.0284
43	10	0.8000	0.6667	0.0000
44	3	0.2000	0.0000	0.0000
44	10	0.8000	0.6667	0.0000

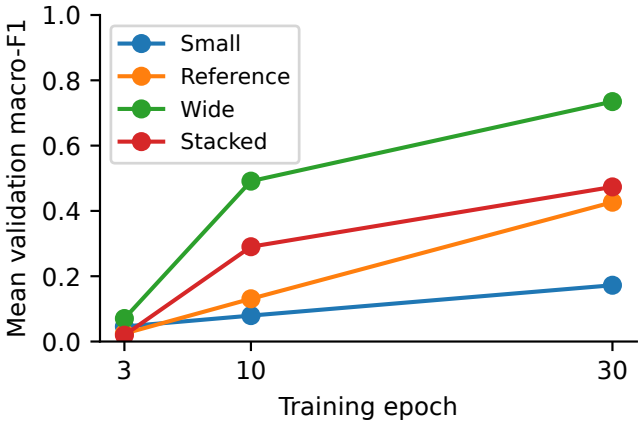


Fig. 2. Validation trajectories for the four presets, averaged over three seeds. These presets were fixed before the runs and were not selected by a completed BPSO search.

TABLE IV
REFERENCE CNN–GRU AT 30 EPOCHS ON THE FROZEN RANDOM SPLIT. MACRO-F1 SHOWS MEAN ± SAMPLE SD; ACCURACY AND FPR SHOW MEANS ACROSS THREE SEEDS.

Scaling	Macro-F1	Accuracy	FPR
Column L2	0.4267 ± 0.2074	0.3136	0.9442
Min–Max	0.7831 ± 0.0768	0.6550	0.5324
Row L2	0.7190 ± 0.0390	0.6710	0.4934

B. Normalization and Component Controls

The reference obtained mean macro-F1 0.4267, 0.7190 and 0.7831 under column L2, row L2 and Min–Max, respectively. The scaling choice affected the measured scores in this experiment. It does not follow that Min–Max is best for other settings, or that a preprocessing gain is evidence of a new search method.

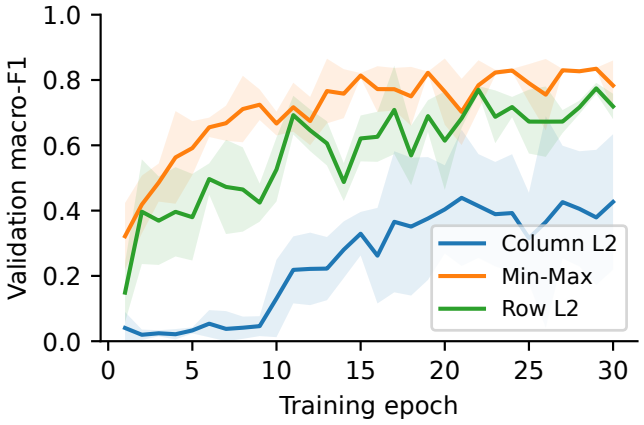


Fig. 3. Reference-model validation curves. Shading represents the sample SD over three training seeds on the same dataset; it is not a confidence interval across datasets.

TABLE V
CONVENTIONAL NEURAL CONTROLS WITH MIN–MAX AT 30 EPOCHS; THREE TRAINING SEEDS. COMPONENT CAPACITIES AND READOUTS DIFFERED.

Model	Macro-F1	Accuracy	FPR	Params.
MLP	0.8325 ± 0.0012	0.7247	0.4217	4,072
CNN	0.8275 ± 0.0032	0.7233	0.4221	7,016
GRU	0.7510 ± 0.0709	0.6416	0.5469	3,624
CNN–GRU	0.7831 ± 0.0768	0.6550	0.5324	5,128

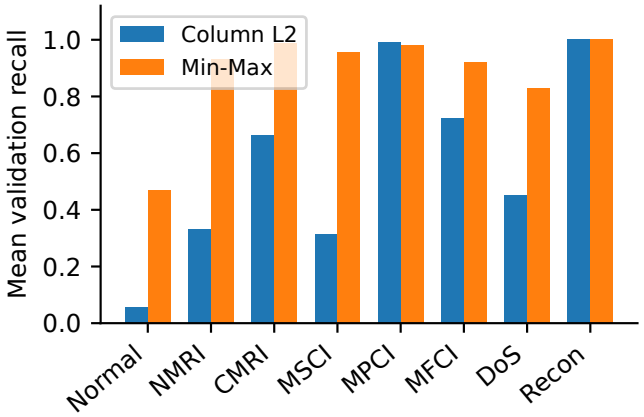


Fig. 4. Mean class recall of the reference at epoch 30 under two scaling choices. These are validation recalls, not new held-out results.

The MLP, CNN-only and GRU-only results are given in Table V. Both the MLP and CNN-only model had higher mean validation macro-F1 than the reference hybrid at this budget. The experiment therefore did not establish a hybrid advantage. We include parameter counts and FPR because model capacity and errors on normal traffic also matter. The Min–Max reference flagged an average of 53.24% of normal validation records as attacks, compared with 42.17% for the MLP. These false-alarm rates do not support deployment readiness. Differences in the remaining model settings also prevent attributing the scores solely to recurrence or convolution.

TABLE VI
SELECTED REFERENCE CONTINUATIONS TO 200 EPOCHS. COLUMN L2 USED THREE SEEDS; MIN–MAX USED SEED 42 ONLY. ALL MEASUREMENTS ARE VALIDATION-ONLY.

Scaling	Seeds	Epochs	Macro-F1	Accuracy
Column L2	3	200	0.8390 ± 0.0084	0.7267
Min–Max	1	200	0.8407 (n=1)	0.7264

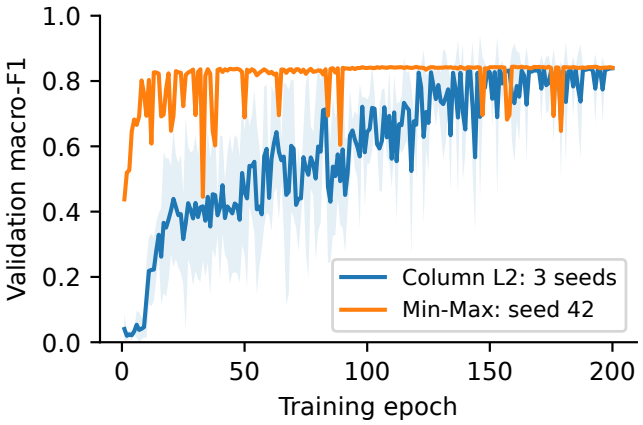


Fig. 5. Long-budget reference trajectories. The Min–Max curve has one seed; its variability is not estimable from this run.

TABLE VII
STATE-GROUPED VALIDATION, MEAN ± SAMPLE SD OVER THREE TRAINING SEEDS. ITS VALIDATION ROWS DIFFER FROM THE RANDOM SPLIT.

Model	Macro-F1	Accuracy	FPR
Random forest	0.8372 ± 0.0010	0.8480	0.1885
Logistic	0.8018 ± 0.0000	0.6806	0.4902
CNN–GRU	0.7559 ± 0.0804	0.6077	0.6158

C. Longer Training and State-Grouped Validation

After 200 epochs, the column-L2 reference had mean validation macro-F1 0.8390, while the one Min–Max continuation reached 0.8407. Scaling therefore affected the comparison differently at different budgets. The unequal seed counts also prevent a general conclusion about which scaling method is best. These runs extend the earlier reference trajectories. The separate final-seed evaluation below supplies fixed-endpoint test measurements; a full-cohort long-budget comparison remains outstanding.

The state-grouped reference achieved mean validation macro-F1 0.7559 (Table VII). This adds evidence under zero exact non-time state-group overlap. It does not establish capture-level independence, prospective detection or cross-release performance. The reserved group partition was not evaluated by a model.

TABLE VIII
FIXED CNN16–GRU32 REFERENCE AFTER 200 EPOCHS WITH MIN–MAX SCALING AND BALANCED LOSS. VALIDATION AND TEST F1 ARE MACRO-AVERAGES. THE HISTORICAL RANDOM-SPLIT TEST PARTITION WAS RETAINED.

Seed	Val. F1	Test acc.	Test F1	FPR
10042	0.8416	0.7242	0.8456	0.4268
10043	0.8423	0.7236	0.8441	0.4271
10044	0.8436	0.7243	0.8465	0.4271

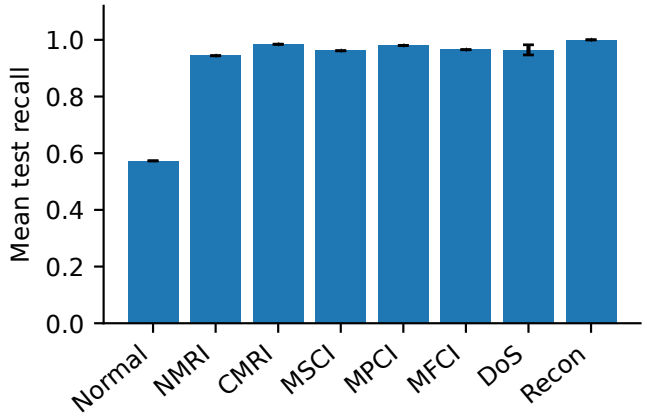


Fig. 6. Test recall at the fixed 200-epoch endpoint. Bars show means and whiskers show sample SD across three final training seeds on the same test rows.

D. Fixed-Endpoint Test Evaluation

The three independent initializations obtained mean test macro-F1 0.8454 $\pm$ 0.0012, mean accuracy 0.7240 and mean normal-class FPR 0.4270 (Table VIII). All three prediction archives reproduced their saved classification metrics and confusion matrices. These results extend the evidence beyond short-budget verification, but they do not identify a BPSO-selected winner or demonstrate a search benefit. The earlier ten-epoch reference used different scaling and training duration, so the difference cannot be attributed to a single factor. Repeating training with different seeds measures training variability. It does not provide independent test data: the runs share test rows, and repeated operating states remain a source of dependence.

Table IX explains the gap between macro-F1 and accuracy. Normal recall was approximately 0.5730, while NMRI precision was approximately 0.0944 despite NMRI recall of 0.9439. About 42.70% of normal records were flagged as attacks, predominantly NMRI. Thus high attack recall coexisted with substantial false alarms, and aggregate macro-F1 alone would overstate practical detector quality. No post-test tuning was performed to reduce these errors.

E. Validation-Selected False-Alarm Correction

The balanced-loss reference was followed by an exploratory comparison of unweighted and square-root inverse-frequency weighting. Architecture, preprocessing, optimizer and the 200-epoch endpoint were held fixed. Candidate training used seed 42 and training/validation arrays only. Neither candidate met the initial 1% validation false-alarm target with at least 90% attack detection. Before generating new test predictions, the false-alarm limit was revised to 3%, retaining the detection constraint. Square-root weighting was selected by validation macro-F1 (0.8552 versus 0.8480 for unweighted loss).

TABLE IX
MEAN FINAL TEST CLASS METRICS ACROSS THREE TRAINING SEEDS. SUPPORT IS THE FIXED TEST CLASS COUNT.

Class	Precision	Recall	F1	Support
Normal	0.9847	0.5730	0.7244	12,231
NMRI	0.0944	0.9439	0.1717	553
CMRI	0.9374	0.9845	0.9604	3,093
MSCI	0.9677	0.9615	0.9646	156
MPCI	0.9810	0.9804	0.9807	1,528
MFCI	1.0000	0.9652	0.9823	115
DoS	0.9944	0.9646	0.9792	367
Recon	1.0000	1.0000	1.0000	1,361

The exploratory validation-selected square-root weighting correction reduced mean normal test false alarms from 42.70% to 1.90%, with accuracy 95.44%, macro-F1 0.8551 and attack detection 90.96%. However, NMRI recall fell from 94.39% to 0.54%. This is a false-alarm versus attack-sensitivity trade-off, not a complete detector-quality solution; the original 1% false-alarm target remains unmet. Independent evaluation is still required. The selected setting was evaluated at paired final seeds 10042, 10043 and 10044, each trained for exactly 200 epochs on GPU. These results describe a conventional loss correction on the development-known split. The revised target is exploratory, and no test-driven threshold or endpoint selection was performed after these new predictions.

F. Dedicated NMRI Specialist

A subsequent validation-selected normal/NMRI specialist increased mean NMRI test recall from 0.54% to 16.82%, while normal false alarms rose from 1.90% to 2.60%. Accuracy was 95.46%, macro-F1 0.8854, and overall attack detection 92.23%. This improves the observed trade-off, but most NMRI attacks are still missed. The same training-fitted specialist was shared across three hybrid seeds; these are not three independent specialist fits or independent-dataset replicates.

The specialist was trained on normal/NMRI training rows and only reconsidered normal or NMRI hybrid predictions. Extra Trees, histogram gradient boosting and a GPU-trained neural specialist were compared on validation. No tested method achieved 50% NMRI recall below 5% false alarms. An explicitly revised exploratory criterion selected histogram gradient boosting by validation macro-F1 under 3% false alarms, at least 10% NMRI recall and 90% overall attack detection. All three validation thresholds were fixed before new test predictions. Training-only analysis found 1,374 of

1,657 NMRI records shared their non-time predictor state with normal records. The audit leaves out duration, so it cannot establish that the full feature set is unable to distinguish the classes.

Setting	FPR (%)	NMRI (%)	Acc. (%)	F1
Balanced	42.70	94.39	72.40	0.8454
Sqrt loss	1.90	0.54	95.44	0.8551
Specialist	2.60	16.82	95.46	0.8854

TABLE X
OBSERVED TEST OPERATING POINTS. NMRI DENOTES CLASS RECALL, AND FPR IS THE NORMAL FALSE-POSITIVE RATE. MEANS USE THREE HYBRID SEEDS; THE SPECIALIST IS SHARED.

G. Historical Evidence and Software Verification

The earlier ten-epoch held-out reference scored 0.299371 macro-F1 and 0.707792 accuracy, with no CMRI or MSCI records detected. We retained those results. The later validation experiments do not supply replacement test scores. Re-scoring saved predictions reproduced the checked metrics for 47 historical files.

The expanded software suite passed 74 tests in each CPU and CUDA-configured run. Component-continuation checks explicitly use CPU; device-aware checks also exercised CUDA. Bounded integration checks exercised preparation, training, search, diagnostics, seven-method comparison, reporting and search resume on real Gas and three synthetic profiles. Tests covered train-fitted transforms, encoding, maximization, resource caps, native scheduler replay and tie-aware rankings. The synthetic results support software execution only. These checks support covered behavior, not every architecture or deployment condition.

VI. COMPUTATIONAL EFFICIENCY OF ADAPTIVE MULTI-FIDELITY BPSO

A. Diagnosis and experimental correction

The earlier matched-wall study gave each method 30 seconds with high iteration ceilings and no target-stop rule. It therefore measured quality at a common budget, not saving to a common quality: BPSO and UA each consumed about 28.76 GPU seconds. Predictor calls cost about 0.10 seconds; its archive cost 403.82 GPU seconds. Continuation and decoded deduplication already worked. Per-example batching, scalar synchronization, dense validation and repeated model/checkpoint loading were execution overheads.

We distinguish fixed-quality and fixed-compute experiments. For target τ, define $T(\tau)$ as cumulative measured GPU elapsed when an actually evaluated 30-epoch candidate first reaches validation macro-F1 τ. Misses are right-censored at budget B; restricted completion time is $\min(T, B)$. Every seed remains in the comparison. Fixed-compute experiments compare best actual 30-epoch scores at common GPU limits and cannot themselves prove cost saving.

GPU elapsed retains the original convention: synchronized training/validation elapsed time, including host stalls within those intervals; it is not pure kernel time or energy. Predictor, scheduling, checkpoint and import overhead also appears in search wall time. GPU caps are checked at completed epoch boundaries, so a last epoch can overrun. Strict-budget score comparisons exclude terminal observations after the stated limit.

B. Engineering and candidate policy

Vectorized batches preserve the original sampler and RNG order. Bounded data/resident-tensor caches reduce loading/transfers, and a one-model cache keyed by trial directory avoids redundant reconstruction and optimizer loading during immediate continuation. FP32 loss-scalar synchronization is deferred. Validation runs at 3/10/30 epochs; checkpoints still commit each epoch. Across three development architectures, cumulative GPU elapsed fell from 25.059 to 20.182 seconds (19.47%), with identical final parameter tensors. Fixed-order one-pass profiles have limited precision. AMP changed outcomes beyond the 0.01 development tolerance and was rejected; batch sizes/search space were unchanged.

The candidate initially probes two particles, then chooses continuations dynamically rather than using a fixed halving quota. A new probe is forced after at most two promotions. Its priority is

$$P(a) = \widehat{f}_{30}(a) + \beta u(a) - \lambda \frac{\widehat{c}(a)}{\operatorname{median}_{z \in \mathcal{A}} \widehat{c}(z)}. \tag{7}$$

where $\widehat{c}$ separates remaining training epochs from remaining validation passes. A candidate is stopped if its full prediction upper bound is below the best observed 30-epoch score plus 0.01; low initial F1 alone does not stop it. Only actual terminal scores update BPSO attractors. All predictive variants retain the same 20% diversity restart.

The sparse, timing-invariant forest is refit using existing logs. Architecture-disjoint development replay on IDs 0–11 fixed $\beta = 0.1$, $\lambda = 0.2$ before the new search seeds. Calibration IDs 12–17 and held-out IDs 18–24 retain the split-conformal residual construction [9]. Nominal 80% intervals covered 13/14 held-out observations at 3 epochs and 14/14 at 10, with mean widths 0.573 and 0.481. These broad intervals do not guarantee useful calibration under adaptive proposal shift. No new GPU archive was collected; historical collection cost is still charged.

C. Measured quality, cost and ablations

Nine methods/ablations used seeds 601–603, the same 8192/4096 training/validation view, all eight classes, original training-fitted Min–Max preprocessing, balanced loss, FP32 runtime and maximum fidelity 30. The initial 8-GPU-second pilot executed independent targets 0.4/0.5/0.6 and fixed-compute runs. Low control attainment motivated a declared exploratory continuation to 20 GPU seconds for target 0.6 and fixed compute. It retained all methods/seeds/coefficients, resumed saved optimizer/RNG states, and charged every previously spent epoch, GPU second and wall second, including

TABLE XI
TARGET 0.6: ACTUAL MEAN GPU ELAPSED TO TARGET OR CAP, WITH ALL THREE SEEDS RETAINED. MISSES ARE CENSORED RATHER THAN EXCLUDED. SCORES ARE ACTUAL TERMINAL VALIDATION MACRO-F1. ARCHIVE COST IS SEPARATE.

Method	GPU s	F1	Saving (%)	Hits
BPSO	17.01	0.264	0.0	1/3
BPSO-SH	12.10	0.497	28.8	2/3
DEHB	6.78	0.703	60.2	3/3
Fixed prediction	11.94	0.529	29.8	2/3
Hyperband	17.88	0.502	-5.1	1/3
Dynamic prediction	12.98	0.507	23.7	2/3
Full UA	12.80	0.539	24.7	2/3
UA without cost	13.03	0.507	23.4	2/3
UA without bonus	12.80	0.539	24.7	2/3

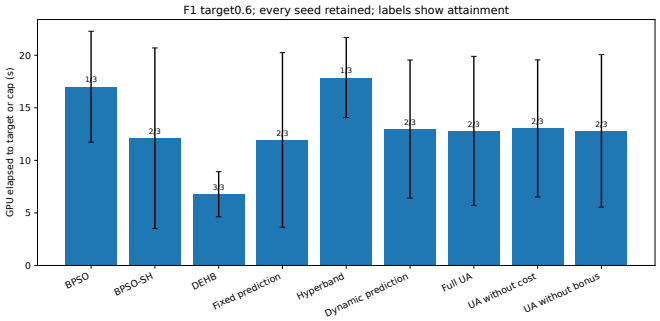


Fig. 7. Actual target-or-cap GPU consumption, including misses. Error bars show search-seed SD; annotations give target attainment.

checkpoint-import overhead. It is not a new independent replication.

Full UA used 12.803 GPU seconds versus 17.008 for BPSO: a 24.72% online reduction, just below the 25% engineering goal. It attained 2/3 targets versus 1/3 for BPSO. SH used 12.103 seconds with 2/3 hits, so full UA was 5.78% slower. DEHB used 6.776 seconds, attained 3/3 targets and saved 60.16% versus BPSO. These finite-horizon, three-seed results do not establish statistical superiority. The restricted-time saving 25.14% is a different censored estimand and must not replace the actual-consumption result to claim the 25% goal was met.

At fixed 20-second compute, mean validation macro-F1 was 0.5391 for full UA,0.2696 for BPSO,0.5030 for SH and 0.7356 for DEHB. The full-method gap to DEHB was 0.1965, exceeding the 0.02 preservation margin. Removing the uncertainty bonus changed no terminal winner scores. Removing cost reduced mean F1 by 0.0320, but cost-aware UA still did not beat SH on target-time cost. Fixed-quota prediction used 11.940 target-or-cap GPU seconds and F1 0.5291; dynamic prediction used 12.978 and 0.5071. The more complex allocation is not independently supported as an improvement.

D. Total cost, final models and limitations

Charging the historical archive gives 416.622 GPU seconds for one full-UA quality search versus 17.008 for BPSO: no net

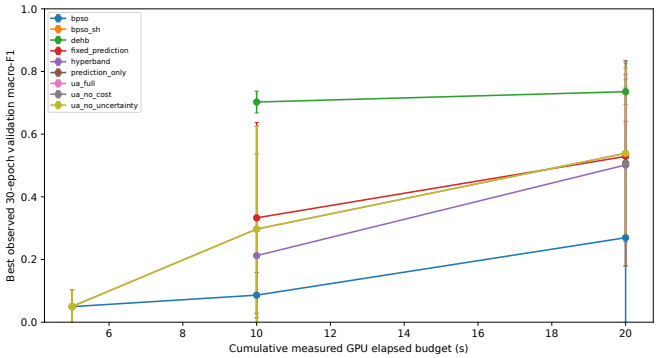


Fig. 8. Fixed-compute quality at 5/10/20 GPU seconds. Means/SD require all three seeds to have a terminal model; missing points are not replaced by zero. Raw seed traces are also retained.

single-search saving remains. The fit/calibration subset alone cost 308.386 historical GPU seconds. Smaller-log CPU refits reduced that attributable subset but increased held-out point-prediction error; they were not substituted into the frozen benchmark. Historical reuse is not free computation. Online calibration was not claimed from a few adaptively selected terminal observations.

All methods' fixed-compute architecture choices were frozen before final testing, using dedicated seed 1601 and 30 epochs. The previously examined historical test is not blind independent data. BPSO test macro-F1 was 0.2733, with normal FPR 0.8038. BPSO-SH test macro-F1 was 0.5759, with normal FPR 0.6180. Full UA test macro-F1 was 0.3100, with normal FPR 0.9442. DEHB test macro-F1 was 0.6859, with normal FPR 0.7549. Precision, attack recalls, confusion matrices, parameters and measured inference latency are retained; one final training seed and shared historical rows limit generalization. Existing conventional MLP/CNN/GRU/reference results remain contextual.

At the preceding study stage, the regression suite passed 105 CPU and 50 targeted CUDA tests. Saved validation/test predictions were re-scored, epoch ledgers were checked, and the old manuscript and results were preserved.

The strongest measured free control is DEHB; SH is the simpler BPSO alternative and beats full UA on target-time cost. Runtime engineering is verified, but novelty and superior uncertainty-aware allocation are not established. The initial pilot and exploratory extension are reported separately; more search/final seeds, full-training studies, independently validated calibration and independent captures remain necessary.

E. Archive-free online extension

The archive-dependent result above motivated a fresh, separately bounded experiment. The new implementation starts with precisely ordinary BPSO–SH proposals, quotas and tie handling. It captures architecture, learning-curve prefix and measured incremental-cost features at epochs 3 and 10, attaching a terminal label only after actual epoch 30 training. No offline bundle, historical target, cross-method checkpoint or

TABLE XII
FRESH TARGET 0.6 RUNS, MEANS ACROSS THREE SEARCH SEEDS. GPU SECONDS ARE MEASURED; ARCHIVE COST IS CHARGED TO A STANDALONE SEARCH. ALL MISSES ARE RETAINED. HITS COUNT ACTUAL TERMINAL TARGETS BEFORE THE STRICT HORIZON.

Method	Online GPU	Archive	Total GPU	F1	Hits
BPSO	16.824	0.00	16.824	0.3054	1/3
BPSO + SH	12.004	0.00	12.004	0.4971	2/3
DEHB	6.778	0.00	6.778	0.7026	3/3
Offline UA	12.631	403.82	416.450	0.5391	2/3
Online prediction	11.919	0.00	11.919	0.4971	2/3
Online + uncertainty	11.950	0.00	11.950	0.4971	2/3
Archive-free full	11.947	0.00	11.947	0.4971	2/3

separately trained GPU archive enters this variant. Candidate maturation is charged to ordinary search work.

A fixed architecture hash assigns fit/calibration/check roles with expected proportions 50/20/30%. CPU random forests use 24 trees of maximum depth 4. All current query architectures are excluded from fitting and calibration. Check forecasts are frozen before their terminal labels arrive. At least six fit, four calibration and four prequential check architectures per rung are required. The point gate requires MAE$\leq$ 0.08, at least 10% and 0.005 absolute MAE improvement over current-F1 prediction, and Spearman$\geq$ 0.3. For uncertainty modes, empirical coverage must be at least 0.75 for nominal 80% intervals, with mean width at most 0.3; current intervals must also meet the width limit. These are conservative engineering checks, not a proof of calibration under adaptive sampling.

The interval uses a separate calibration quantile of absolute residual divided by forest dispersion (floor 0.02). If useful, ranking uses the earlier priority equation with generic frozen $\beta = 0.1$, $\lambda = 0.05$ and median-normalized measured remaining cost. Prediction-only sets both coefficients to zero; uncertainty-only sets $\lambda = 0$. Failure of a gate retains ordinary SH ranking. Only actual terminal observations update swarm bests. Fixed SH quotas are intentionally retained to isolate ranking; no new BPSO equation or algorithmic novelty is claimed.

F. Fresh end-to-end comparison

The 42 new runs use seeds 601–603, seven methods, target 0.6 and fixed 20-GPU-second phases. Dataset rows, eight labels, preprocessing, architecture space, FP32 execution and 3/10/30 fidelities match the preceding 20-second experiment. A 50-second wall safety cap also applies. Method order is a seeded permutation; methods run serially. Earlier checkpoints are not imported. Misses are retained and censored; strict-horizon quality excludes terminal evaluations beyond the cap, while actual resource consumption includes epoch-boundary overruns. Times follow the existing synchronized elapsed convention.

Archive-free full consumed 11.947 GPU seconds versus 16.824 for BPSO, a 28.99% total reduction, with 2/3 versus 1/3 target attainment. Terminal validation macro-F1 means were 0.4971 and 0.3054 respectively. SH consumed 12.004 seconds and produced identical winner scores to the archive-free variants. The small 0.47% timing difference does not establish an advantage over SH. DEHB used 6.778 seconds and reached 3/3 targets. Removing the archive lowers standalone total cost relative to offline UA (416.450 seconds), but the online component itself must earn its cost.

TABLE XIII
FRESH FIXED 20-GPU-SECOND COMPARISON. F1 IS BEST GENUINELY EVALUATED 30-EPOCH VALIDATION MACRO-F1, MEAN (SAMPLE SD). EPOCH COUNTS AND ACTUAL GPU ELAPSED INCLUDE CHARGED INCREMENTAL WORK.

Method	Terminal F1 (SD)	Epochs	GPU s
BPSO	0.2696 (0.3715)	178.7	20.040
BPSO + SH	0.5030 (0.3231)	138.3	20.088
DEHB	0.7356 (0.0402)	195.0	20.044
Offline UA	0.5391 (0.2719)	205.0	20.201
Online prediction	0.5030 (0.3231)	137.3	20.086
Online + uncertainty	0.5030 (0.3231)	137.3	20.089
Archive-free full	0.5030 (0.3231)	137.7	20.080

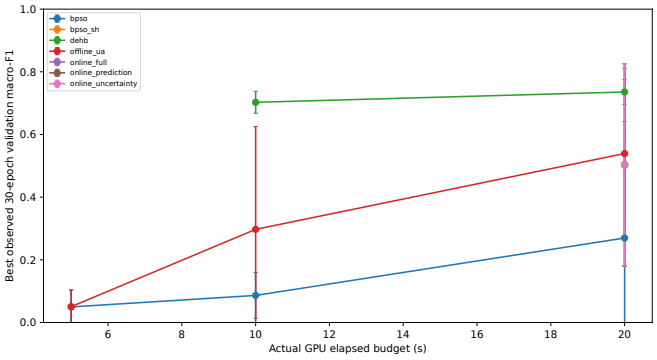


Fig. 9. Fresh fixed-compute anytime quality. Error bars show search-seed SD; points require all three terminal scores. Equal budget consumption is not evidence of cost saving.

Across the 18 matched online trials, predictive promotion activated 0 times and CPU fitting ran 0 times. Thus the new modes effectively executed SH in this short-budget study. Their ablations do not establish any benefit from uncertainty or cost ranking. The zero external archive cost is real; an online-learning advantage is not demonstrated. CPU preparation ranged 0.0017–0.0078 seconds per run and is included in search wall time.

At fixed 20-second compute, archive-free full preserved SH scores but trailed offline UA by 0.0361 and DEHB by 0.2325 absolute macro-F1. Both exceed the 0.02 engineering preservation margin; competitive quality against the strongest control was not maintained.

The full archive is charged once for a reusable multi-search campaign in the accompanying CSV, rather than once per seed. Offline CPU development preparation is not fully isolated; recorded archive wall time 420.722 seconds therefore supplies only a lower bound for full offline wall preparation. Online CPU fitting/inference and scheduler costs are recorded separately. All saved validation predictions were independently

re-scored, and epoch/GPU ledgers and architecture-disjoint feature/predictor identities were verified. The expanded suite passed 110 CPU and 55 targeted CUDA-configured tests.

The separate seed 601 diagnostic used 120.061 GPU seconds and 128.449 wall seconds, matured 16 distinct terminal architectures, performed 1 CPU fits and applied prediction at 0 promotion decisions. Preparation consumed 0.0475 CPU wall seconds. Gate counts: insufficient fit or calibration architectures: 32; insufficient out of time check architectures: 1. This is a cold-start diagnostic, not a matched performance comparison.

No new final detector test was executed in this extension. The preceding historical test results are retained, and the test remains development-known. The evidence supports removing an uneconomic archive and using SH fallback; it does not support superior online uncertainty-aware allocation. DEHB remains the strongest measured control, with SH the simpler BPSO alternative. More seeds, a prespecified larger matched horizon, useful online calibration and independent data remain necessary before an IEEE optimizer contribution can be defended.

VII. Limitations and Remaining Research

The new measurements reduce several evidence gaps, but an adaptive-prediction advantage remains unestablished. Full-scale BPSO, random search, Hyperband and DEHB comparisons with matched actual resources are still incomplete. The four-preset study and bounded 25-architecture pilot do not cover full-training search reliability, a complete three-seed search study or a complete 200-epoch cohort. The preset rankings were measured under column L2 and may change with another normalization. A mean score, a learning curve or a successful command check is not enough to establish statistical superiority or speedup.

The source paper leaves uncertainty about its splits, averaging, feature order and layer conventions. Our recurrent blocks process feature positions within a record, not a sequence of captured events. The normalization and component experiments cover one release and only a few seeds. Grouping removed exact shared predictor states, but capture identities remain unknown and the class proportions changed. The dataset owner's warnings about operating conditions [10], [8] still apply. We have not evaluated the original ToN-IoT/Bot-IoT/SWaT releases, validated deployment, measured energy savings or established equivalence across multiple GPUs.

The next experiments should use a fixed protocol, capture metadata or an independent release, matched-resource search controls and a locked final evaluation endpoint. Any proposed allocation rule also needs comparison with relevant prior methods and ablations of its components. Those questions require additional measurements.

The new efficiency study adds explicit target stopping, actual GPU-budget comparisons and an exploratory charged continuation. Its controls, engineering profiles, censoring and archive-inclusive costs are reported in Section VI; none establish superior uncertainty-aware optimization.

VIII. Reproducibility

Each experiment retains the dataset checksum, predictor order, partition identities, preprocessing parameters and training settings. We fitted imputation, scaling and class weights on training data only. The validation runs used the same candidate identities for seeds 42, 43 and 44. Final reference training restarted at seeds 10042, 10043 and 10044, with evaluation after exactly 200 epochs. Model weights, optimizer state and random-number states were saved in checkpoints so higher budgets could continue earlier runs. Re-scoring the saved predictions verified the metrics and confusion matrices. The means and sample standard deviations describe training variability on fixed partitions; assessing broader generalization requires independent releases.

IX. Conclusion

The implemented framework lets us examine hybrid intrusion-detection models under a training budget. The experiments show that training duration and normalization affect candidate fitness. At 30 epochs, reference validation macro-F1 was 0.7831 under Min–Max and 0.4267 under column L2. Across all three seeds in the four-preset cohort, the winner at epoch-10 was also the winner at epoch 30. This gives a reason to check the training setup before attributing weak early scores to the search method.

The independently initialized reference models reached mean test macro-F1 0.8454 and sample standard deviation 0.0012 at the fixed 200-epoch endpoint. These scores belong to a fixed hybrid evaluated on the existing split, rather than to a BPSO-selected winner. Per-class errors, false alarms and repeated predictor states still limit the result. The project provides a runnable framework and measurements of training-budget sensitivity; establishing an adaptive-prediction advantage and deployment performance requires larger matched-resource comparisons and independent evaluation data.

The archive-free follow-up removed separately trained GPU predictor observations and reduced total target-or-cap GPU cost by 28.99% relative to BPSO. It remained in SH fallback during the short matched searches, so this saving does not establish a contribution from online prediction or uncertainty. DEHB was the stronger measured control. The contribution is reproducible engineering and a candid optimizer comparison; a new IEEE optimizer-superiority claim is not supported.

The exploratory square-root weighting correction, selected on validation data, lowered mean normal test false alarms from 42.70% to 1.90%. Accuracy was 95.44%, macro-F1 was 0.8551 and attack detection was 90.96%, but NMRI recall dropped from 94.39% to 0.54%. This is a false-alarm versus attack-sensitivity trade-off, not a complete detector-quality solution; the original 1% false-alarm target remains unmet. Independent evaluation is still required.

A subsequent validation-selected normal/NMRI specialist increased mean NMRI test recall from 0.54% to 16.82%, while normal false alarms rose from 1.90% to 2.60%. Accuracy was 95.46%, macro-F1 0.8854, and overall attack detection 92.23%. This improves the observed trade-off, but most NMRI

attacks are still missed. The same training-fitted specialist was shared across three hybrid seeds; these are not three independent specialist fits or independent-dataset replicates.

REFERENCES

[1] K.-D. Lu, Y.-W. Yang, G.-Q. Zeng, C. Peng, G.-G. Geng, and J. Weng, "BPSO-AHDL-IDS: Binary particle swarm optimization-based automated hybrid deep learning model for intrusion detection of internet of things," *IEEE Transactions on Automation Science and Engineering*, vol. 22, pp. 15 859–15 877, 2025.

[2] J. Bergstra and Y. Bengio, "Random search for hyper-parameter optimization," *Journal of Machine Learning Research*, vol. 13, pp. 281–305, 2012. [Online]. Available: https://www.jmlr.org/papers/v13/bergstra12a.html

[3] K. Jamieson and A. Talwalkar, "Non-stochastic best arm identification and hyperparameter optimization," in *Proceedings of the 19th International Conference on Artificial Intelligence and Statistics*, ser. Proceedings of Machine Learning Research, vol. 51, 2016, pp. 240–248. [Online]. Available: https://proceedings.mlr.press/v51/jamieson16.html

[4] L. Li, K. Jamieson, G. DeSalvo, A. Rostamizadeh, and A. Talwalkar, "Hyperband: A novel bandit-based approach to hyperparameter optimization," *Journal of Machine Learning Research*, vol. 18, no. 185, pp. 1–52, 2018. [Online]. Available: https://www.jmlr.org/papers/v18/16-558.html

[5] N. Awad, N. Mallik, and F. Hutter, "DEHB: Evolutionary hyberband for scalable, robust and efficient hyperparameter optimization," in *Proceedings of the Thirtieth International Joint Conference on Artificial Intelligence*, 2021, pp. 2147–2153. [Online]. Available: https://ijcai.org/proceedings/2021/296

[6] S. Falkner, A. Klein, and F. Hutter, "BOHB: Robust and efficient hyperparameter optimization at scale," in *Proceedings of the 35th International Conference on Machine Learning*, ser. Proceedings of Machine Learning Research, vol. 80, 2018, pp. 1437–1446. [Online]. Available: https://proceedings.mlr.press/v80/falkner18a.html

[7] T. Morris and W. Gao, "Industrial control system traffic data sets for intrusion detection research," in *Critical Infrastructure Protection VIII*, ser. IFIP Advances in Information and Communication Technology, J. Butts and S. Shenoi, Eds. Springer, 2014, vol. 441, pp. 65–78. [Online]. Available: https://link.springer.com/chapter/10.1007/978-3-662-45355-1_5

[8] J. Hsu, D. Mudd, and Z. Thornton, "Project report—SCADA anomaly detection," Mississippi State University, Tech. Rep., 2014, project period February 17–April 30, 2014; Table 1, PDF pp. 2–3. [Online]. Available: http://www.ece.uah.edu/~thm0009/icsdatasets/MSU_SCADA_Final_Report.pdf

[9] A. N. Angelopoulos and S. Bates, "A gentle introduction to conformal prediction and distribution-free uncertainty quantification," 2021. [Online]. Available: https://arxiv.org/abs/2107.07511

[10] T. Morris, "Industrial control system (ICS) cyber attack datasets," Author's dataset catalogue, accessed November 7, 2026. Distinguishes the older Gas/Water release from New Gas Pipeline. [Online]. Available: https://sites.google.com/a/uah.edu/tommy-morris-uah/ics-data-sets